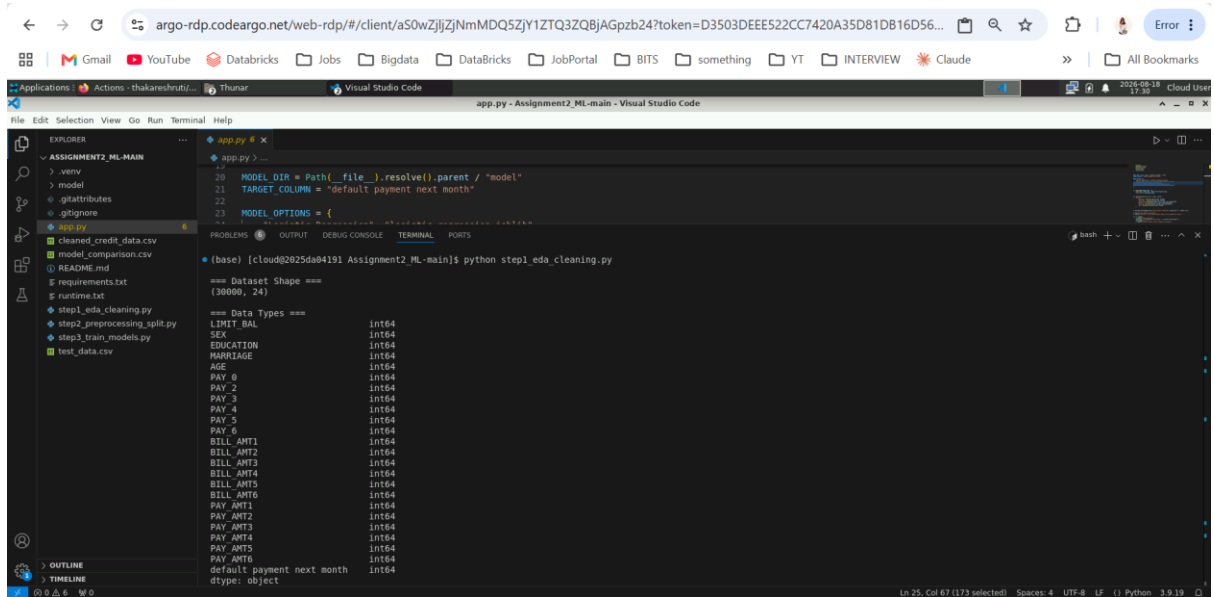
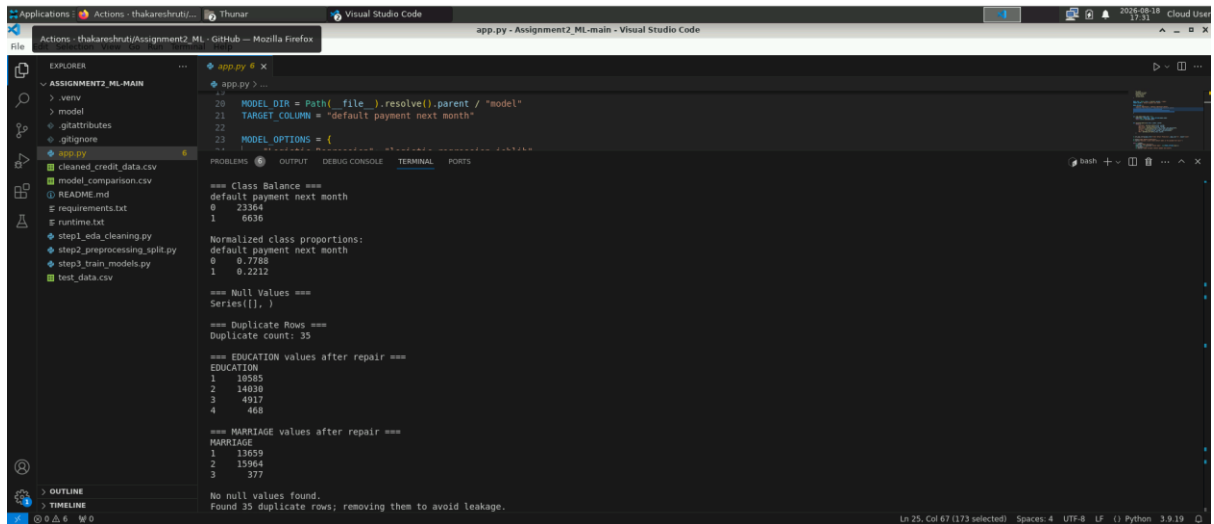


Screenshots of Output



```
argop-rdpcodeargo.net/web-rdp/#/client/as0wZjIjZjNmMDQ5ZjY1ZTZQ3ZQBjAGpzb247token=D3503DEEE522CC7420A35D81DB16D56...
Applications: Actions: thakareshruti... Thunar Visual Studio Code
app.py - Assignment2_ML-main - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
  ASSIGNMENT2_ML-MAIN
    .venv
    model
    .gitattributes
    .gitignore
    app.py
    cleaned_credit_data.csv
    model_comparison.csv
    README.md
    requirements.txt
    runtime.txt
    step1_eda_cleaning.py
    step2_preprocessing_split.py
    step3_train_models.py
    test_data.csv
  OUTLINE
  TIMELINE
  (base) [cloud02025d0b4191 Assignment2_ML-main]$ python step1_eda_cleaning.py
===== Dataset Shape =====
(30000, 24)

===== Data Types =====
LIMIT_BAL      int64
SEX            int64
EDUCATION      int64
MARRIAGE       int64
AGE            int64
PAY_0          int64
PAY_2          int64
PAY_3          int64
PAY_4          int64
PAY_5          int64
PAY_6          int64
BILL_AMT1      int64
BILL_AMT2      int64
BILL_AMT3      int64
BILL_AMT4      int64
BILL_AMT5      int64
BILL_AMT6      int64
PAY_AMT1       int64
PAY_AMT2       int64
PAY_AMT3       int64
PAY_AMT4       int64
PAY_AMT5       int64
PAY_AMT6       int64
default payment next month  int64
dtype: object
```



```
Applications: Actions: thakareshruti... Thunar Visual Studio Code
app.py - Assignment2_ML-main - Visual Studio Code
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EXPLORER
  ASSIGNMENT2_ML-MAIN
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    step2_preprocessing_split.py
    step3_train_models.py
    test_data.csv
  OUTLINE
  TIMELINE
  (base) [cloud02025d0b4191 Assignment2_ML-main]$ python step2_preprocessing_split.py
===== Class Balance =====
default payment next month
0    23364
1     6636

Normalized class proportions:
default payment next month
0    0.7788
1    0.2212

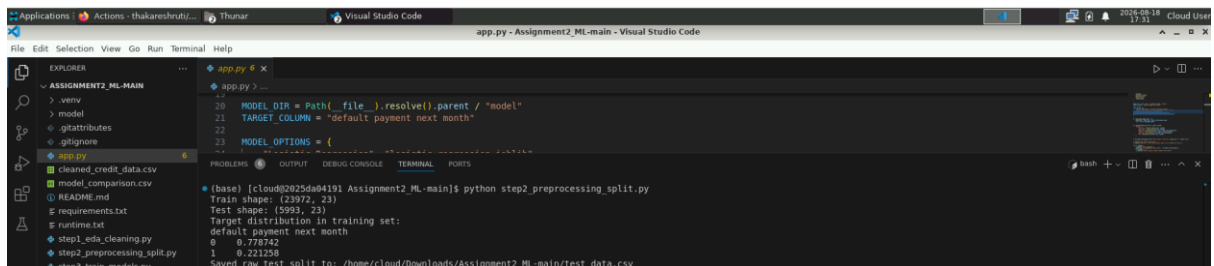
===== Null Values =====
Series([], dtype=object)

===== Duplicate Rows =====
Duplicate count: 35

===== EDUCATION values after repair =====
EDUCATION
1    10585
2    14030
3     4917
4     468

===== MARRIAGE values after repair =====
MARRIAGE
1    13659
2    13964
3     377

No null values found.
Found 35 duplicate rows; removing them to avoid leakage.
```



```
Applications: Actions: thakareshruti... Thunar Visual Studio Code
app.py - Assignment2_ML-main - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
  ASSIGNMENT2_ML-MAIN
    .venv
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    step2_preprocessing_split.py
    step3_train_models.py
  OUTLINE
  TIMELINE
  (base) [cloud02025d0b4191 Assignment2_ML-main]$ python step3_train_models.py
Train shape: (23972, 23)
Test shape: (5993, 23)
Target distribution in training set:
default payment next month
0    0.778742
1    0.221258
Saved raw test split to: /home/cloud/Downloads/Assignment2_ML-main/test_data.csv
```

```
File Edit Selection View Go Run Terminal Help
app.py - Assignment2_ML-main - Visual Studio Code

EXPLORER
  ASSIGNMENT2_ML-MAIN
    .venv
    model
    .gitattributes
    .gitignore
    app.py
    cleaned_credit_data.csv
    model_comparison.csv
    README.md
    requirements.txt
    runtime.txt
    step1_eda_cleaning.py
    step2_preprocessing_split.py
    step3_train_models.py
    test_data.csv

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Saved raw test split to: /home/cloud/Downloads/Assignment2_ML-main/test_data.csv
(base) [cloud@2025da04191 Assignment2_ML-main]$ python step3_train_models.py

=== Logistic Regression ===
Accuracy AUC Score Precision Recall F1 Score MCC
0.77674 0.76282 0.496032 0.565611 0.528541 0.384518

=== Decision Tree Classifier ===
Accuracy AUC Score Precision Recall F1 Score MCC
0.729351 0.603568 0.385978 0.377823 0.38186 0.286851

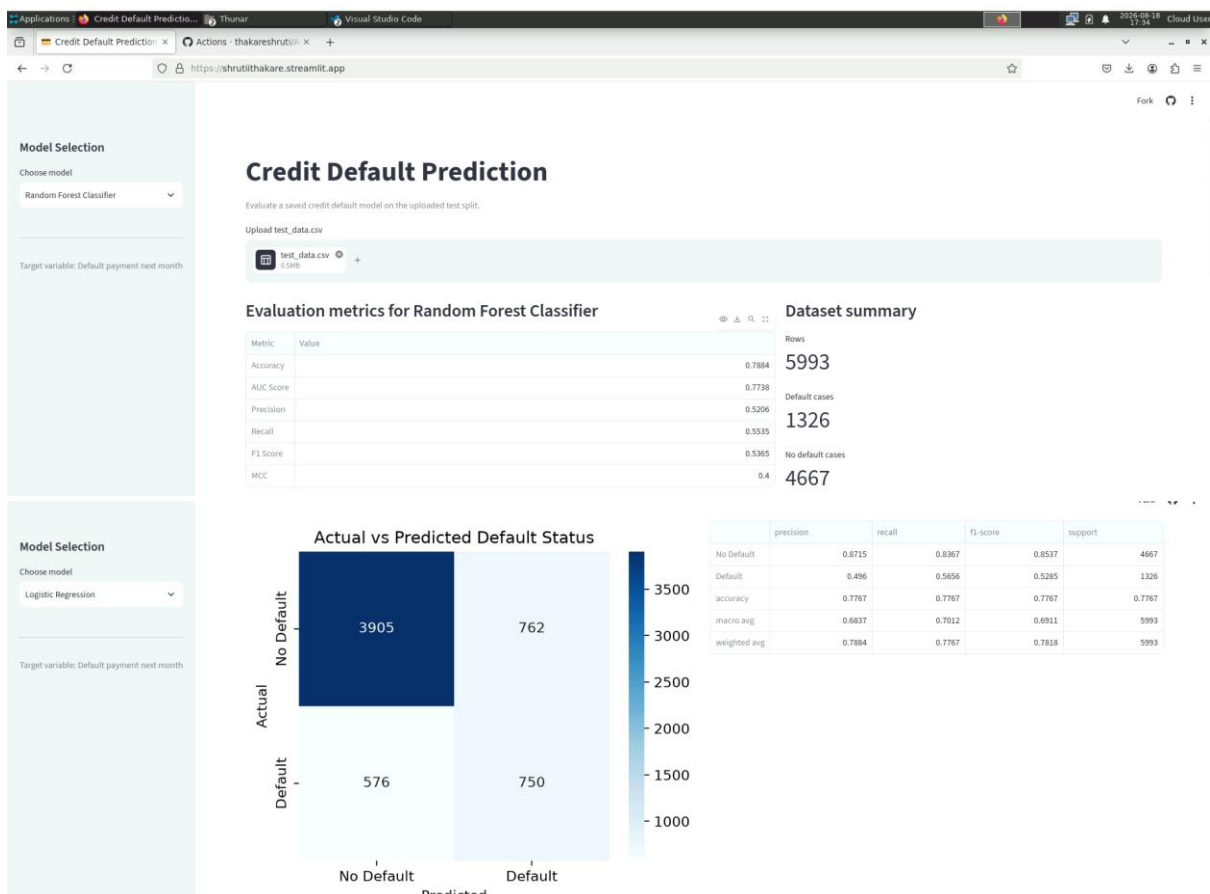
=== KNN Classifier ===
Accuracy AUC Score Precision Recall F1 Score MCC
0.795894 0.6949 0.561713 0.33635 0.420755 0.320526

=== Gaussian Naive Bayes ===
Accuracy AUC Score Precision Recall F1 Score MCC
0.796429 0.733653 0.653179 0.170437 0.270335 0.257565

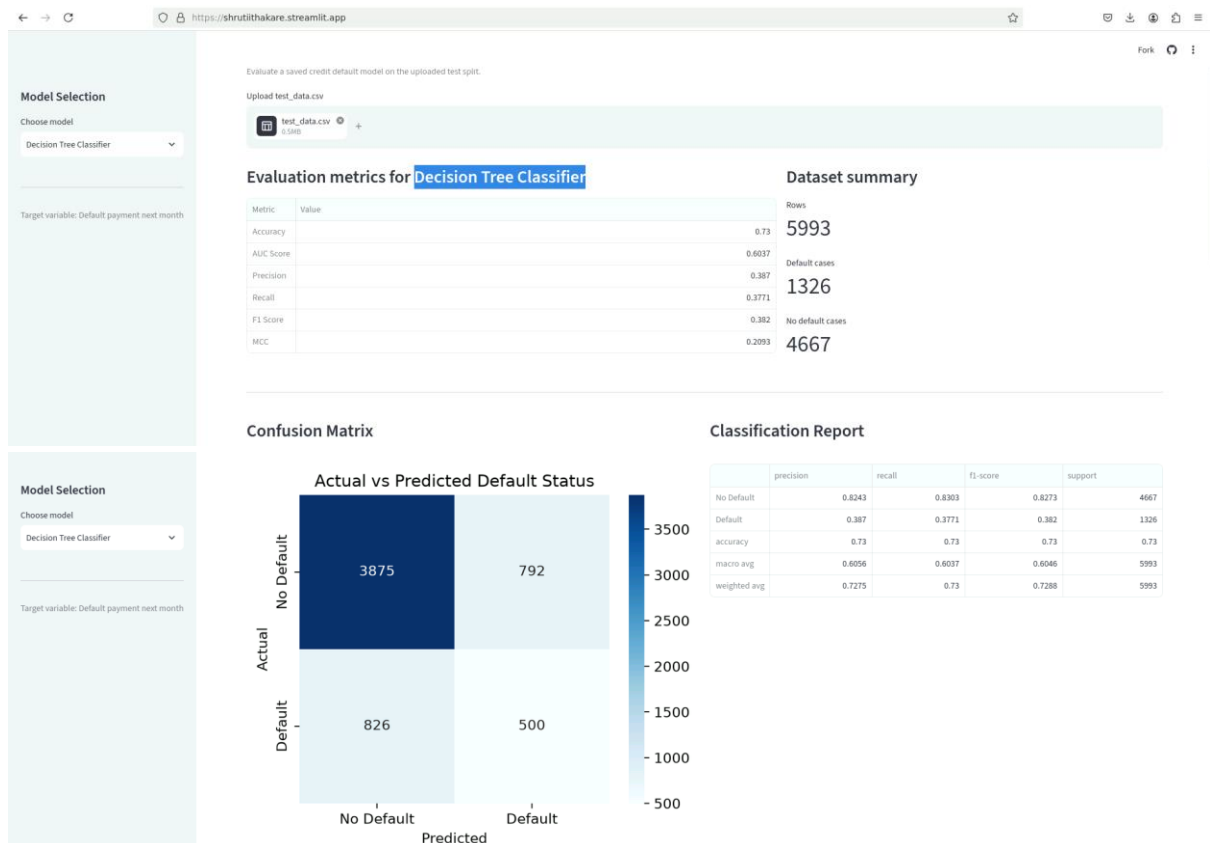
=== Random Forest Classifier ===
Accuracy AUC Score Precision Recall F1 Score MCC
0.75972 0.770236 0.466352 0.595777 0.523179 0.370646

=== Full model comparison ===
Accuracy AUC Score Precision Recall F1 Score MCC
Model
Logistic Regression 0.7767 0.7629 0.4960 0.5656 0.5285 0.3845
Decision Tree Classifier 0.7294 0.6036 0.3860 0.3778 0.3819 0.2867
KNN Classifier 0.7951 0.6949 0.5617 0.3363 0.4208 0.3205
Gaussian Naive Bayes 0.7964 0.7337 0.6532 0.1704 0.2703 0.2576
Random Forest Classifier 0.7597 0.7702 0.4664 0.5958 0.5232 0.3706
(base) [cloud@2025da04191 Assignment2_ML-main]$ python step1_eda_cleaning.py
```

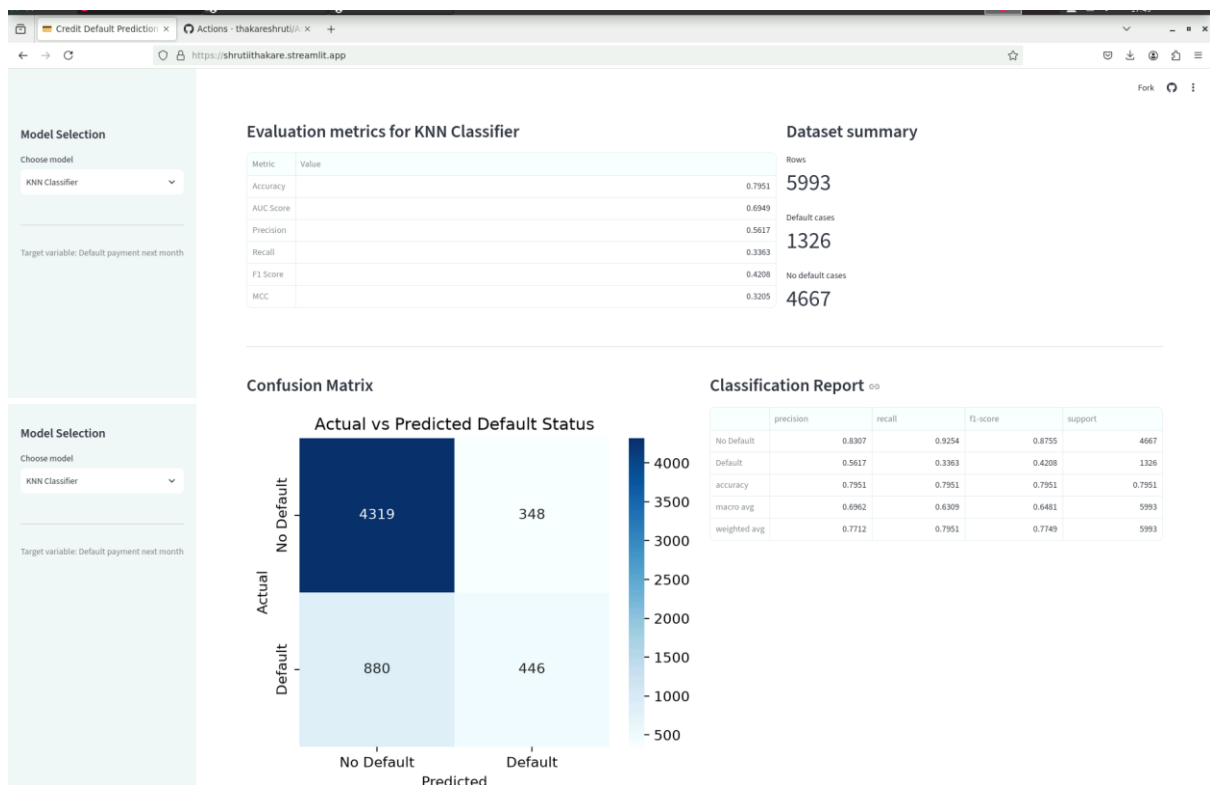
Decision Tree Classifier



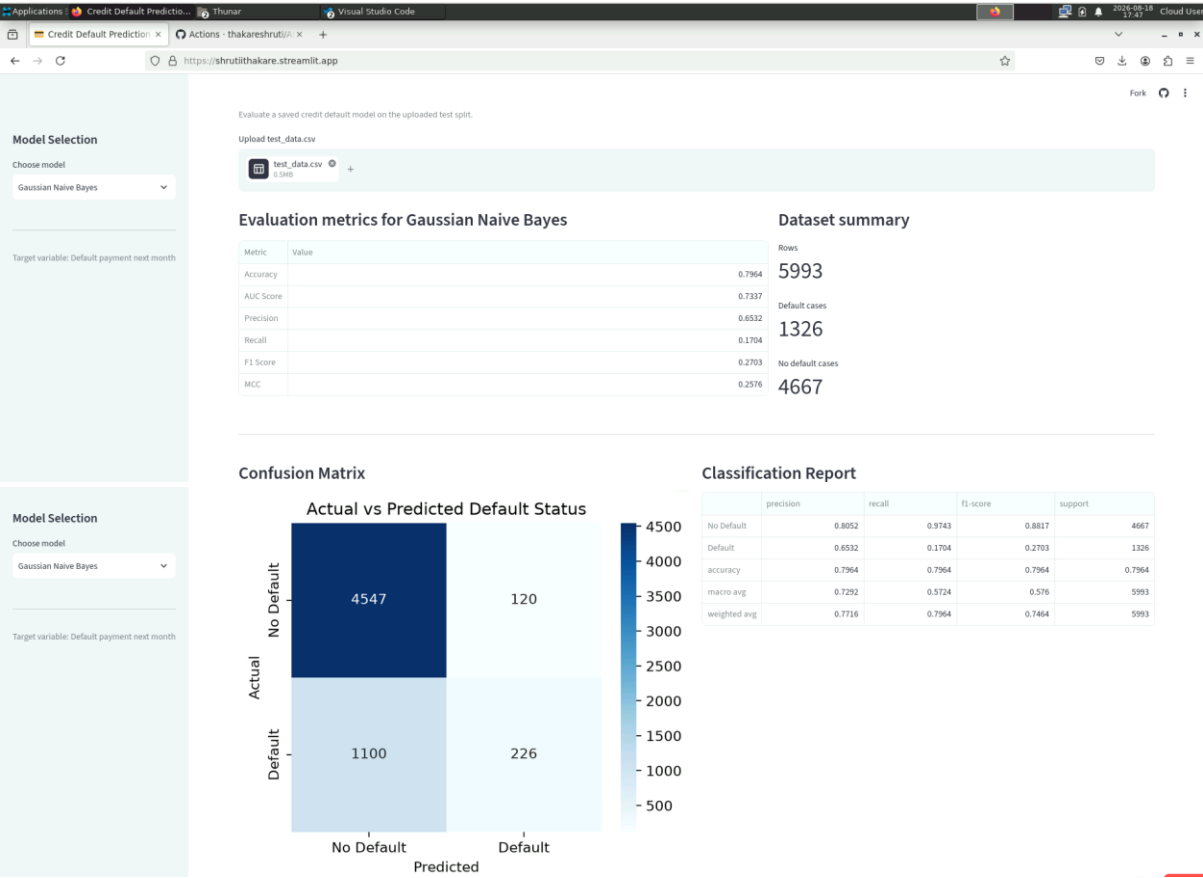
Logistic Regression



KNN Classifier



Gaussian Naive Bayes



Random Forest Classifier

